

Exp No: 9

ULTRASONIC SENSOR IN PROTEUS USING ARDUINO

PROGRAM:

// Program Sketch

```
int trig = 10; int echo = 9; long duration; int cm; void setup() { //  
put your setup code here, to run once: pinMode(trig, OUTPUT);  
pinMode(echo, INPUT); Serial.begin(9600); } void loop() { // Sketch  
to calculate ultrasonic distance digitalWrite(trig, LOW);  
delayMicroseconds(10); digitalWrite(trig, HIGH);  
delayMicroseconds(10); digitalWrite(trig, LOW);  
delayMicroseconds(10); duration = pulseIn(echo, HIGH); cm =  
(duration/2) * 0.034; // Print Value on Serial Monitor Serial.print("  
Distance = "); Serial.print(cm); Serial.println(" cm"); }PROGRAM:
```

// Program Sketch

```
int trig = 10;  
  
int echo = 9;  
  
long duration;  
  
int cm;  
  
void setup() {  
// put your setup code here, to run once:  
pinMode(trig, OUTPUT);  
  
pinMode(echo, INPUT);  
  
Serial.begin(9600);  
  
}
```

```

void loop() { // Sketch to calculate ultrasonic distance

digitalWrite(trig, LOW);

delayMicroseconds(10);

digitalWrite(trig, HIGH);

delayMicroseconds(10);

digitalWrite(trig, LOW);

delayMicroseconds(10);

duration = pulseIn(echo, HIGH);

cm = (duration/2) * 0.034;

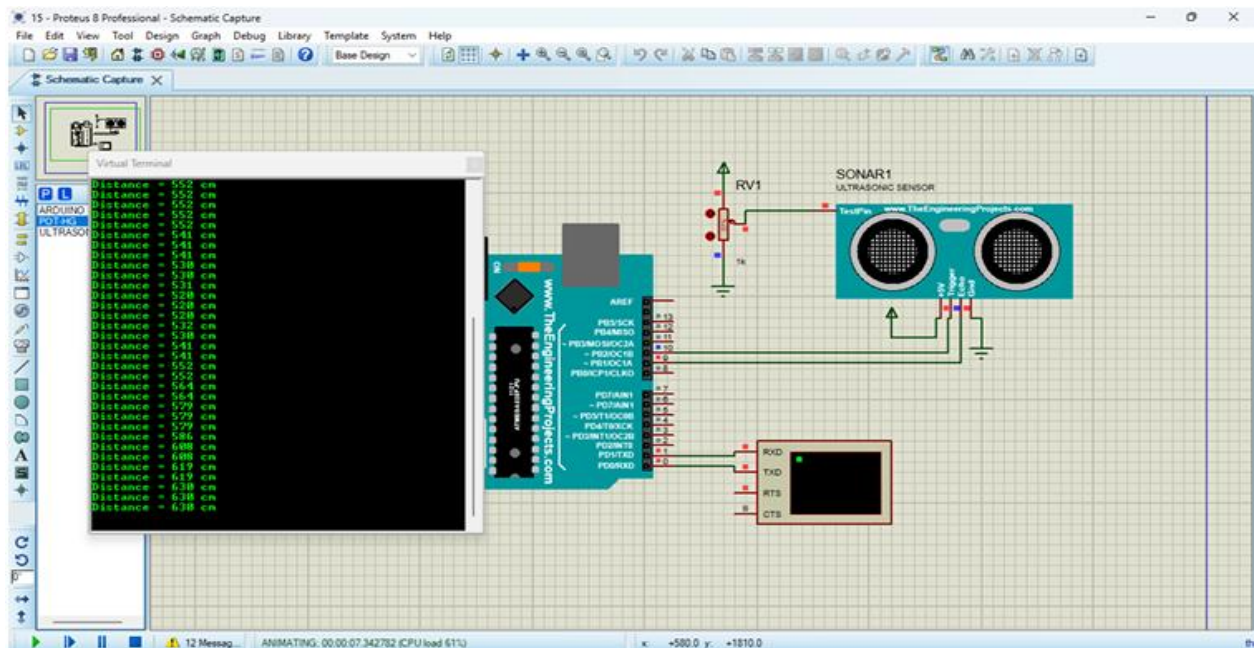
// Print Value on Serial Monitor

Serial.print(" Distance = ");

Serial.print(cm);

Serial.println(" cm");

```



```

}

```